

Quantized Llada Evaluation

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Outline

- Solved Memory Issues related to using pytorch instead of gemlite
- Evaluated SING quantized Llada Model on some benchmarks

Memory Issues

When quantizing and evaluating llada model I used the `sing_nogemlite` method

[MEM] after model load	alloc=4.457 GB reserved=4.475 GB peak=4.457 GB
[MEM] generate() start	alloc=4.457 GB reserved=4.475 GB peak=4.457 GB
[MEM] block 0 start	alloc=4.457 GB reserved=4.475 GB peak=4.457 GB
[MEM] block 0 step 000 post-forward	alloc=4.507 GB reserved=9.357 GB peak=9.324 GB



Solution: Offloading unused weights and scales to CPU

Use the offload function used for gemlite inference to reduce memory when using pytorch inference

```
def _offload_originals_post_pack(self):
    # Offload quantized weights and meta to CPU; runtime only needs GemLite buffers.
    if self.W_q is not None and self.W_q.is_cuda:
        self.W_q = self.W_q.cpu()
    if isinstance(self.meta, dict):
        for k, v in list(self.meta.items()):
            if isinstance(v, torch.Tensor) and v.is_cuda:
                self.meta[k] = v.cpu()
    torch.cuda.empty_cache()
```

Solution: Offloading unused weights and scales to CPU

[MEM] after model load	gpu_alloc=0.965 GB gpu_reserved=0.967 GB gpu_peak=1.208 GB
[MEM] generate() start	gpu_alloc=0.965 GB gpu_reserved=0.967 GB gpu_peak=1.208 GB
[MEM] block 0 start	gpu_alloc=0.965 GB gpu_reserved=0.967 GB gpu_peak=1.208 GB
[MEM] block 0 step 000 post-forward	gpu_alloc=1.017 GB gpu_reserved=7.674 GB gpu_peak=6.076 GB

Llada Model Evaluation on WikiText/Wiki2

- 128 Monte carlo estimates is used when evaluating on Wiki-text data
- 1 monte carlo estimate is used when evaluating on Wiki2 data
- The numbers reported are word perplexity (the lower the better)

	Wiki text	Wiki2
Llada-8B-Base	5.21	5.44
SINQ-Llada-4-bit	5.32	5.55
Perplexity Increase	2.1%	1.98%

Method	Qwen3-1.7B			Qwen3-14B			Qwen3-32B		
	Mem.	Wiki2↓	C4↓	Mem.	Wiki2↓	C4↓	Mem.	Wiki2↓	C4↓
Original (BF16)	3.44	16.67	19.21	29.54	8.64	12.01	65.52	7.60	10.77
3-BIT	RTN [†]	1.28	32.43	31.10	9.23	10.50	14.88	17.61	30.78
	Hadamard + RTN [†]	1.28	32.40	31.07	9.23	10.60	15.10	17.61	11.26
	HQQ	1.28	32.10	30.54	9.23	10.73	14.39	17.62	9.09
	SINQ (ours)	1.28	22.39	24.88	9.25	9.33	12.90	17.61	8.79
4-BIT	RTN [†]	1.42	18.74	20.81	10.54	8.95	12.50	20.78	8.92
	Hadamard + RTN [†]	1.42	19.10	20.70	10.54	8.85	12.35	20.78	8.28
	HQQ	1.42	18.96	22.10	10.54	8.78	12.36	20.78	8.62
	SINQ (ours)	1.42	17.14	19.83	10.56	8.76	12.21	20.73	7.74

Llada Model evaluation on Ceval Data

- CEval is a chinese benchmark consisting of multiple choice questions from 52 subjects
- Numbers reported are accuracy values

	Ceval
Llada-8B-Base	70.0
SINQ-Llada-4-bit	68.5
Accuracy Drop	2.14%

	LLaDA 8B*	LLaMA3 8B*	LLaMA2 7B*	Qwen2 7B [†]	Qwen2.5 7B [†]	Mistral 7B [†]	Deepseek 7B [¶]
Model	Diffusion	AR	AR	AR	AR	AR	AR
Training tokens	2.3T	15T	2T	7T	18T	-	2T
General Tasks							
MMLU	65.9 (5)	65.4 (5)	45.9 (5)	70.3 (5)	74.2 (5)	64.2 (5)	48.2 (5)
BBH	49.7 (3)	62.1 (3)	39.4 (3)	62.3 (3)	70.4 (3)	56.1 (3)	39.5 (3)
ARC-C	45.9 (0)	53.1 (0)	46.3 (0)	60.6 (25)	63.7 (25)	60.0 (25)	48.1 (0)
Hellaswag	70.5 (0)	79.1 (0)	76.0 (0)	80.7 (10)	80.2 (10)	83.3 (10)	75.4 (0)
TruthfulQA	46.1 (0)	44.0 (0)	39.0 (0)	54.2 (0)	56.4 (0)	42.2 (0)	-
WinoGrande	74.8 (5)	77.3 (5)	72.5 (5)	77.0 (5)	75.9 (5)	78.4 (5)	70.5 (0)
PIQA	73.6 (0)	80.6 (0)	79.1 (0)	-	-	-	79.2 (0)
Mathematics & Science							
GSM8K	70.3 (4)	48.7 (4)	13.1 (4)	80.2 (4)	85.4 (4)	36.2 (4)	17.4 (8)
Math	31.4 (4)	16.0 (4)	4.3 (4)	43.5 (4)	49.8 (4)	10.2 (4)	6.0 (4)
GPQA	25.2 (5)	25.9 (5)	25.7 (5)	30.8 (5)	36.4 (5)	24.7 (5)	-
Code							
HumanEval	35.4 (0)	34.8 (0)	12.8 (0)	51.2 (0)	57.9 (0)	29.3 (0)	26.2 (0)
HumanEval-FIM	73.8 (2)	73.3 (2)	26.9 (2)	-	-	-	-
MBPP	40.0 (4)	48.8 (4)	23.2 (4)	64.2 (0)	74.9 (0)	51.1 (0)	39.0 (3)
Chinese							
CMMLU	69.9 (5)	50.7 (5)	32.5 (5)	83.9 (5)	-	-	47.2 (5)
C-Eval	70.5 (5)	51.7 (5)	34.0 (5)	83.2 (5)	-	-	45.0 (5)

Plan for Next Week

- Do more evaluation on quantized Llada Model
- Quantize Dream Model and evaluate it on similar benchmarks